

A FIELD GUIDE TO DIRECT TO GARMENT PRINTING

DTG PRINTING

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Direct-to-garment printing puts water-based pigment ink directly into fabric. That one fact generates every rule in this guide — about color, durability, file prep, and the real cost of making things bright. This guide explains the science, flags the failure modes, and gives you the tools to make deliberate decisions instead of accidental ones.

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Compiled by Mumblejinx · v1.1 · July 2026

The science here is drawn from publicly available materials research and manufacturer documentation — factual information that belongs to no one. Both color palettes were independently derived by Mumblejinx from that science. The synthesis, framing, and tools are original work.

A NOTE ON THIS GUIDE

This is researched, not tested. I've pulled together the best information I could find on DTG printing and tried to be careful with it, but I haven't been able to verify these results on my own equipment. Your inks, garments, pretreatment,

and printer will differ from whatever a given source assumed — so verify anything that costs money against your own setup, and test on scrap before a full run. DTG involves real materials and real money; treat this as a well-researched starting point, not a manufacturer's spec. If something here is wrong, I'd like to know so I can correct it.

BEFORE ANYTHING ELSE

THE THIRTY-SECOND VERSION

Flat fill, hard edge — always.

Gradients and soft alpha edges deposit continuously varying ink volumes that the binder system can't handle uniformly — and they're the most common cause of cracking. Flat fills with hard edges cost you nothing stylistically and gain you real durability.

Know your ink ceiling.

Total ink coverage (the sum of all channels) runs roughly 200–240% on light shirts and 280–320% on dark shirts once the white underbase is included. Stay under your RIP software's limit for your specific printer.

Light garment beats dark garment.

A white or light shirt needs no white underbase — only your design's own ink deposits. That's the single largest durability lever available, and it costs nothing at the design stage.

100% cotton, 5–6oz.

Cotton's hydroxyl-rich surface bonds well with pigment binders. Polyester repels water-based ink and, at curing temperatures, can bleed its own dye into white ink. Past ~50% poly, results degrade noticeably.

Your screen is lying to you.

DTG's CMYK+white gamut is narrower than sRGB. Pure blues print violet-shifted. True neons are physically unreproducible. Use a soft-proof ICC profile to see what will

Vibrancy costs wash-life.

More saturated, heavier-coverage colors deposit more pigment binder — which cracks and washes out faster. Choose how you spend that budget deliberately, not by

actually print before you commit.

accident.

01 / THE CHEMISTRY

WHAT DTG INK ACTUALLY IS — AND HOW IT HOLDS

DTG inks are water-based pigment dispersions: microscopic pigment particles suspended in water, not dissolved dyes. Understanding this distinction is the foundation of everything else in this guide — because the way those pigment particles attach to fabric determines how long they stay there.

PIGMENT VS. DYE — AND WHY IT MATTERS FOR DURABILITY

A common shorthand describes DTG ink as sinking "into" the fiber. This is true in one sense — ink doesn't sit as a thick film on top the way screen-print plastisol does — but it's imprecise in an important way. DTG pigment particles don't chemically bond to cotton fiber at a molecular level the way reactive dyes do in garment dyeing. Instead, they're held *at* the fiber surface by a polymer binder that cures around and adheres to both the pigment and the cotton. Think of it as millions of tiny particles locked into a flexible adhesive that grips the weave — not ink soaked into the fiber itself.

This is why curing temperature and dwell time matter so much: you're not just driving off water, you're cross-linking a binder polymer around the pigment. Get that wrong and the binder doesn't set — and the pigment washes away with it, regardless of how good the pigment itself is. It's also why pretreatment chemistry affects adhesion so directly, as covered in [Section 05](#).

The practical upshot: DTG prints do have a soft, low-profile hand-feel compared to screen print because there's no thick plastisol film sitting on top of the weave. But the mechanism is binder adhesion at the fiber surface — not molecular penetration into the fiber.

WHAT ACTUALLY HAPPENS DURING A PRINT RUN

Here's the sequence, step by step:

1. **RIP processing.** A Raster Image Processor (RIP) converts your file into printer head commands, separates channels, handles color management, and enforces ink limits. The RIP — not your design application — decides the final ink volumes on each pixel.
2. **Pretreatment (dark garments only).** A cationic pretreatment solution is applied to the fabric. This changes the fiber's surface charge, allowing the white ink to coagulate near the surface rather than wicking deep into the weave. Without it, white ink would simply disappear into dark fibers. Light garments may need little or no pretreatment.
3. **White underbase (dark garments only).** White pigment ink lays down first as an opaque "ground." Color ink printed onto dark fiber without this base would be invisible — the fiber's own color overwhelms it completely.
4. **Color passes.** Cyan, magenta, yellow, and black inks print in registered passes on top of the white base (or directly onto pretreated light fabric).
5. **Heat curing.** The garment passes through a conveyor dryer (production) or under a heat press (small batch) at temperatures typically between 300–340°F (149–171°C) depending on printer and ink system. Heat drives off the water carrier and cross-links the polymer binder, locking pigment to fiber. Temperature and dwell time work together — see [Section 06](#).

Steps 2, 3, and part of 5 only apply to dark garments. That's not a footnote — it's the single biggest variable in this guide.

02 / FAILURE MODES

SIX WAYS A PRINT ACTUALLY FAILS

"It didn't last" covers a lot of ground. In practice, a DTG print fails in one of six specific ways — and each one points back to a specific, identifiable cause in either the artwork, the production process, or the garment itself.

Cracking

Cause: excessive total ink coverage in a single area. The cured binder film becomes thicker than the fabric beneath it can flex with — and repeated washing and drying eventually fractures it.

Most vulnerable: heavy gradients, large dark solid fills, anywhere white underbase and saturated color stack. When total coverage exceeds your printer's

Fading

Cause: under-curing (binder didn't fully cross-link), accumulated UV exposure, or harsh washing over time.

Most vulnerable: light tones, thin ink deposits, and halftone areas — there's simply less binder-held pigment to absorb the loss before it becomes visible.

recommended limit, cracking is likely, not possible.

Washout

Cause: surface of a thick ink deposit reaches cure temperature while its core remains under-cured. It looks finished. It isn't, underneath — and the first wash removes the un-bonded binder.

Most vulnerable: heavy ink zones, and the white underbase on dark garments where the deposit is thickest.

Color Shift

Cause: DTG's CMYK+white gamut is narrower than the sRGB gamut your screen displays. The RIP maps out-of-gamut colors to the nearest reproducible equivalent — which is never the same as the original.

Most vulnerable: saturated blues (violet-shift is a documented CMYK characteristic), bright greens, and anything neon. Soft-proofing with an ICC profile before printing is the only reliable way to see this in advance.

Crocking

Cause: excess binder sitting on the fiber surface rubs off when the print contacts another surface — another garment, upholstery, skin. Wet crocking (when wet fabric rubs against something) is more severe than dry crocking and is tested by industry standard AATCC Test Method 8.

Most vulnerable: dark, heavy fills on dark garments where the combined underbase + color deposit is thickest. Also: prints that weren't fully cured.

Fibrillation

Cause: over time and with repeated washing, cotton fibers break through the cured ink layer, creating a "vintage" or worn appearance. This is intrinsic to the pigment-on-fiber system — it happens eventually on all DTG prints.

Most visible: dark or high-contrast areas on light garments, and designs with fine detail. Some designers embrace it deliberately; others mitigate it by choosing slightly lower-contrast palettes.

WHAT A SHIRT THAT SURVIVES TEN YEARS ACTUALLY HAS GOING FOR IT

The exact same file, printed on two different shirts, can produce wildly different results in color, durability, and feel. Fiber content, fabric weight, and garment color are production variables as consequential as anything in the artwork itself.

Fiber: cotton bonds best.

Cotton's surface hydroxyl groups interact well with the polymer binders in DTG ink, giving them something to grip. Cotton also absorbs pretreatment fluid evenly, which is critical for white ink adhesion on dark garments. Ringspun cotton gives slightly smoother print surface than open-end spun.

Weight: heavier holds up better.

A 5.5–6oz+ fabric has more fiber mass to anchor the binder system and stays dimensionally stable under the heat of curing. Lighter fabrics flex more under the same stress and give the binder less to hold onto — contributing to faster cracking and fibrillation.

Polyester: dye migration is the real problem.

It's not just that polyester "resists" water-based ink. Poly-cotton blends introduce a specific failure mode called *dye migration*: at curing temperatures, the disperse dyes used to color the polyester fibers can sublimate and bleed into the white ink layer, causing a color cast that wasn't in the design. This is most visible on dark or brightly colored poly-cotton garments. Past ~50% polyester, dye migration becomes a significant risk; past ~80%, DTG generally isn't the right method — DTF (direct-to-film) or screen print handle poly blends better.

Garment color: the biggest lever you control.

A white or light garment needs little to no white underbase — only your design's own ink deposits. A dark garment requires a full white underbase pass just to make color visible at all. That underbase is a substantial additional ink deposit with its own coverage budget and curing requirements. Choosing a light garment for a given design is the single highest-impact durability decision you can make at the artwork stage — it costs nothing and removes the underbase variable entirely.

YOUR SCREEN AND YOUR PRINTER AREN'T SPEAKING THE SAME LANGUAGE

Your file is RGB. Your printer is CMYK plus white. The RIP software handles the translation — but how well it does depends heavily on what you give it and how you've told it to interpret color. This is the most controllable stage in the entire process.

PPI, RESOLUTION, AND LINE WEIGHT — THE NUMBERS THAT MATTER

PPI, not DPI. When talking about your file's resolution, the correct term is PPI (pixels per inch) — the number of pixels in your file per inch of output size. DPI (dots per inch) refers to the physical output resolution of the printer itself, which is a separate and much higher number. In practice, most DTG artists use "DPI" loosely to mean file resolution, but knowing the difference matters when a printer or RIP asks for specific settings: they mean PPI for your file.

Resolution targets. 150 PPI at final print size is the workable floor for most designs. 300 PPI is the standard target that gives the RIP enough information to work with cleanly. Past 300–600 PPI there's diminishing returns — the RIP resamples the file on its way to the printer, and the print heads themselves have a fixed physical resolution. Submitting at 1200 PPI doesn't produce a sharper result; it just creates a larger file and slower processing.

Line weight. 2pt at 300 PPI is a practical minimum for any line meant to survive printing. 3–4pt is the durable, reliable target. Lines thinner than this deposit too little binder to hold reliably to the fiber — they're often the first element to wash out, well before any other part of the design shows wear.

ICC PROFILES — THE MOST SKIPPED STEP IN DTG COLOR MANAGEMENT

An ICC (International Color Consortium) profile is a file that precisely describes how a specific printer, ink set, and substrate combination translates color. Your RIP software uses it to map your RGB design into the ink values that will produce the closest available result. Most professional DTG setups ship with or can download printer-specific ICC profiles from the manufacturer (Epson, Brother, Kornit, etc.).

The practical value for designers is **soft proofing**: with the right ICC profile loaded in Photoshop (View → Proof Colors → Custom → select your printer's profile), you can see what your design will actually look like when printed, before anything goes near a machine. Colors that fall outside the printable gamut will be highlighted — you can adjust them at the design stage rather than discovering the problem on the finished shirt.

If you're printing through a service (Threadless, Printful, etc.) rather than on your own printer, ask what ICC profile they recommend for soft proofing. Using the generic sRGB or "Generic CMYK" profile produces soft proofs that may bear little resemblance to the actual output. Using their specific profile is the only way to soft-proof with any accuracy.

GAMUT, COLOR SHIFT, AND METAMERISM

05 / PRETREATMENT

PRETREATMENT IS DOING CHEMISTRY, NOT PLUGGING HOLES

Pretreatment fluid is often described as "sealing" the fabric — as if it were blocking pores in a sponge. The real mechanism is more interesting, and understanding it explains why so many dark-garment problems trace back to this step.

WHAT PRETREATMENT IS ACTUALLY DOING

Raw cotton fiber has a naturally negative (anionic) surface charge. DTG pigment dispersions and their binder systems are also negatively charged — which means on untreated cotton, like charges repel and the ink wicks deep into the weave rather than staying near the surface. On a white or light garment this is manageable. On a dark garment where you need white ink to form an opaque, surface-level layer, it's catastrophic.

Pretreatment fluids typically contain **cationic (positively charged) polymers** — often polyamines or similar compounds. When applied to the fabric and dried, they reverse the fiber surface's charge from negative to positive. The negatively charged pigment particles and binder in the arriving ink are electrostatically attracted to that positive surface and coagulate near it rather than wicking deeper into the weave. That coagulation is also what produces the solid, opaque white layer — the white ink essentially precipitates at the fiber surface rather than dispersing through it.

This is a charge interaction, not a mechanical seal. The fiber isn't "plugged" — its chemistry has been temporarily altered to attract and hold the ink where it's needed.

WHERE PRETREATMENT GOES WRONG — AND WHAT IT DOES TO CURING TEMPERATURES

06 / CURING

CURED IS A TEMPERATURE, A CLOCK, AND A SYSTEM — NOT A GLANCE

A print can look perfectly finished coming off the dryer and still fail its first wash. Curing is where the binder polymer cross-links around the pigment particles and bonds them to the fiber. Get it wrong and everything else you did right doesn't matter.

TEMPERATURE, DWELL TIME, AND WHY THEY WORK TOGETHER

There is no single universal DTG curing temperature. Different ink systems from different manufacturers cure at different temperatures and require different dwell times — and using the wrong specification for your specific printer and ink combination produces under-cured prints even if you're hitting the target perfectly. Always follow your ink manufacturer's published cure specifications, not generic DTG guidelines.

That said, the typical operating range across most water-based DTG pigment ink systems is **300–340°F (149–171°C)** for a dwell time of 60–90 seconds in a conveyor dryer, or 45–60 seconds under a heat press. Some systems, including certain Kornit configurations, cure differently. The numbers on your dryer's control panel are not the temperature the ink actually experiences — calibrate with independent measurement tools (a donut probe in a conveyor, heat test strips on a press) because equipment drifts.

Temperature and dwell time have a compensatory relationship up to a point: a slightly lower temperature with longer dwell time can achieve the same cure as a higher temperature with shorter dwell — but there are limits in both directions. Too low a temperature never fully cross-links the binder regardless of dwell time. Too high a temperature can scorch the fabric or degrade the pigment before the binder has time to cure evenly.

The failure mode to watch for is **surface cure masking core under-cure**: a thick ink deposit (especially a white underbase) can reach cure temperature on its outer surface while the center remains liquid. It looks and feels finished. But the first wash removes the un-bonded material beneath the surface layer. The only reliable test is a wash test — not a visual check, not a scratch

test, not a tape test.

One practical routine worth adopting: wait 24–72 hours after printing before the first wash. The binder continues to complete its cross-linking at ambient temperature after heat curing, and giving it that window before a full detergent wash gives it the best chance to set completely.

07 / COLOR VS. DURABILITY

VIBRANCY AND LONGEVITY ARE SPENT FROM THE SAME BUDGET

Total ink coverage is driven by three things: how saturated a color is, how dark or light it sits, and how many ink channels stack at any one point. A loud, saturated palette laid down as heavy fills deposits more binder-held pigment per square inch than a muted one — and more deposit per square inch is exactly the condition that causes cracking, washout, and crocking.

The two reference palettes in this guide sit at opposite ends of that trade deliberately. The "DTG Safe" palette publishes its own engineering constraints right on the chart: saturation capped at 65%, lightness held between 22–76%, and no single RGB channel above 80%. Those three thresholds are a measurable, checkable hedge against excessive ink load — not arbitrary taste. The "Safest Colors" vibrant palette relaxes all three, trading some of that durability margin for liveliness.

That gives us something better than a style preference: an actual, checkable rule. The tool below applies it to any color.

THE TOOL

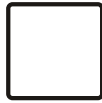
THE INK-LOAD CHECKER

Type a hex code or pick a color. This checks it against the same three thresholds that built the durable palette below — saturation, lightness, and single-channel maximum — and gives you a real verdict, not a vibe.

Methodology note: "single-channel max" is read here as the highest of red, green, or blue expressed as a percent of 255 — the most direct reading of that published rule. Because verdicts are computed fresh from exact pixel values rather than copied off the chart, a few of the very palest swatches below land a hair past their own stated cutoff. That's the tool doing its job, not the palette being wrong — treat it as a close, transparent approximation rather

than a guaranteed bit-for-bit match to the original chart's internal math.

#C31760



CAUTION

Exceeds one threshold. Fine as an accent; riskier as a large, heavy fill.

SATURATION	79%
LIGHTNESS	43%
SINGLE-CHANNEL MAX	76%

PASS

Inside all three thresholds. Built for heavy fills, dark garments, and a long wash life.

CAUTION

Exceeds one threshold. Fine as an accent or on a light garment — riskier as a large, heavy fill.

VIBRANCY MODE

Exceeds two or more thresholds. Expect faster fade or cracking on a heavy fill — spend it deliberately.

THERE

The most useful move isn't "always durable" or "always vibrant" — it's knowing which one a specific design actually needs, and spending the budget accordingly.

Reach for the durable end when —

- it's a flagship piece meant to be worn and washed for years
- it's going on a dark garment, where the underbase already doubles the ink load
- it's a large-area fill rather than a small accent

The vivid end is completely fine when —

- it's a limited run, not meant to outlive the drop
- it's a small accent shape, not the bulk of the fill
- day-one vibrancy matters more than wash #40 for this piece

FOUR HYBRID MOVES THAT GET YOU BOTH

1. **Color-budget your fills.** Spend saturation on small, high-impact areas — a logo mark, an outline, one accent shape — and build the bulk of the large fill area from the durable end of the spectrum. Visual energy reads from contrast and placement, not from every pixel being maximally saturated.
2. **Print on light garments whenever the design allows it.** Removing the white underbase is worth more "vibrancy budget" than almost any single color decision, since it removes roughly half the deposited ink for the same visual result.
3. **Keep every fill flat with a hard edge, regardless of which palette you're drawing from.** This is the one rule that's entirely free — it costs zero vibrancy and buys back real wash-life on either end of the spectrum.
4. **Route the truly electric colors to paper, not fabric.** Posters, stickers, and other paper products can hold a neon that DTG physically cannot reproduce anyway. Reserving the loudest colors for a medium where they can actually show up costs you nothing on the apparel side.

There's a third option beyond "pick the durable palette" or "pick the vibrant palette" for a single shape — and it doesn't break the one rule that's actually free. Halftone screening lets two fully flat, fully durable ink layers sit side by side at a small enough scale that the eye blends them into a single in-between color, without ever printing a gradient.

WHY DOTS AREN'T A LOOPHOLE IN THE FLAT-FILL RULE

Every dot in a halftone pattern is still its own small, flat, hard-edged shape, printed at full strength in exactly one color. Nothing is fading into anything else at the ink level — you're building a tone the same way a CMYK print already builds every color you've ever seen come out of an inkjet: by placing tiny patches of a few flat colors close enough together that the eye averages them into one. This guide's earlier rule was never really "no halftones" — it was "no gradients and no soft alpha edges," because *those* deposit a continuously varying amount of ink that the printer has to commit to at full resolution. A halftone dot is the opposite: a deliberate, controlled, 100%-or-0% decision repeated many times across a shape.

That control is exactly what makes it useful here. At any single point on the garment, the ink is either the durable color or the vibrant one — never both stacked on top of each other the way a transparency blend would stack them. Across the whole shape, the *average* ink demand lands somewhere between the two, in proportion to how much area each color's dots cover.

WHERE THIS TRADES ONE RISK FOR A DIFFERENT ONE

DURABLE ANCHOR

VIBRANT ACCENT

#AC4841



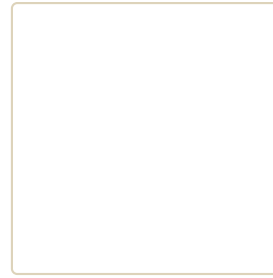
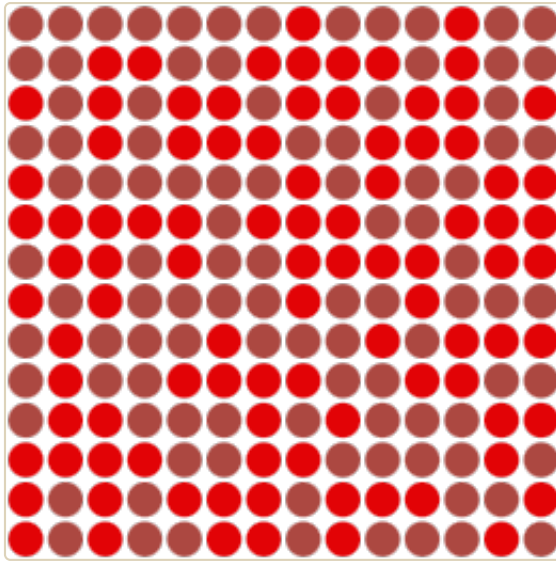
#E20306



Dot coverage

50% vibrant accent





If these were flatly mixed as a single literal color — reference only. This is *not* how the halftone actually prints; your eye does the blending above.

CAUTION

Exceeds one threshold. Fine as an accent; riskier as a large, heavy fill.

SATURATION	69%
LIGHTNESS	46%
SINGLE-CHANNEL MAX	78%

REFERENCE

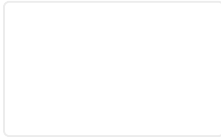
BOTH PALETTES, IN FULL

Every swatch below carries its own verdict, computed live by the same rule as the checker tool above. Click any swatch to load it there directly.

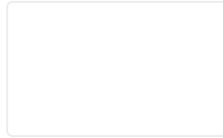
DURABLE — ENGINEERED FOR MAXIMUM WASH-LIFE

Values as specified on the original reference chart: saturation $\leq 65\%$, lightness between 22–76%, no channel above 80%.

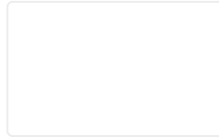
REDS



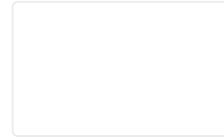
Deep Brick
#802626



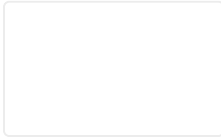
Brick Red
#AC4841



Muted Red
#C46962

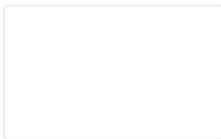


Dusty Rose
#D2948C

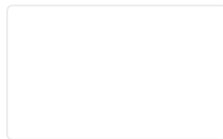


Blush
#E1B6B0

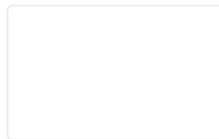
ORANGES



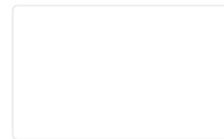
Burnt Sienna
#9E4E20



Terra Cotta
#C07044

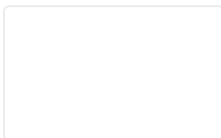


Warm Peach
#D49B6E

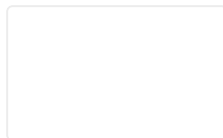


Light Peach
#E4BE9E

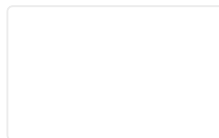
YELLOWS & GOLDS



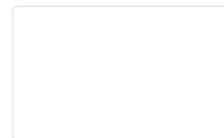
Ochre
#987626



Mustard Gold
#C5A23A

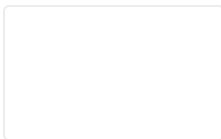


Golden Yellow
#E1C15A

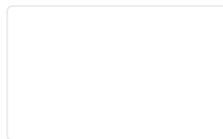


Pale Yellow
#F1E1A6

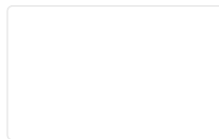
GREENS



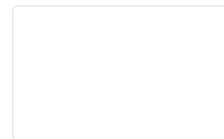
Deep Forest
#2E5E3E



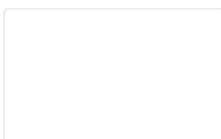
Pine Green
#447C59



Sage Green
#6E9A7B

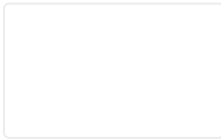


Moss Green
#8DAF8F

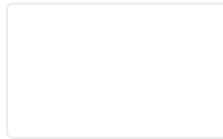


Soft Mint
#B7D5C1

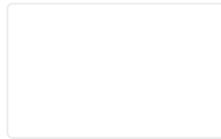
TEALS & TURQUOISES



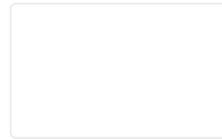
Deep Teal
#1F6B6D



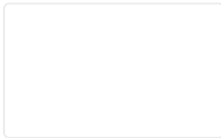
Teal
#30908E



Turquoise
#4FB6B1



Aqua
#7CCDC7



Seafoam
#A6E1DB

BLUES



Deep Navy
#1E3A5F



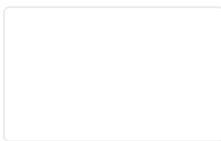
Royal Blue
#2D5EA6



Steel Blue
#4D7DB6

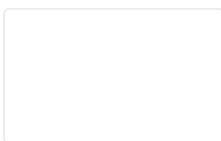


Sky Blue
#7FB3D6

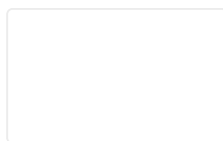


Light Blue
#A9CDEA

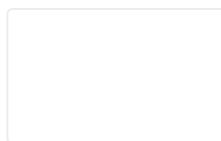
PURPLES



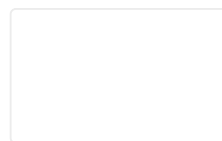
Deep Purple
#4B2F7F



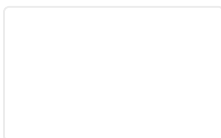
Plum
#6B4EA0



Violet
#8367B8

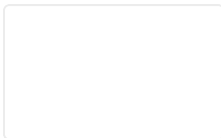


Lavender
#A58FD0

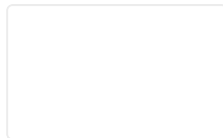


Lilac
#C8B4E2

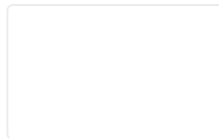
PINKS



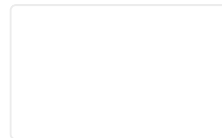
Raspberry
#B43A6F



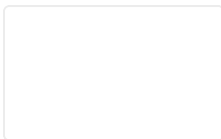
Fuchsia
#C94A92



Rose Pink
#D970A5

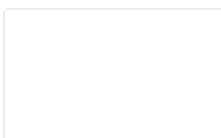


Soft Pink
#E7A7C6

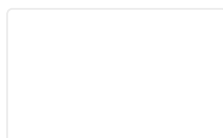


Blush Pink
#F2C6D3

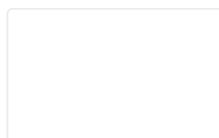
BROWNS



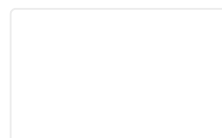
Dark Brown
#4A2E1F



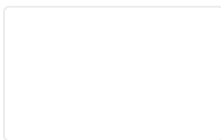
Chocolate
#6B442E



Chestnut
#8A5A3A

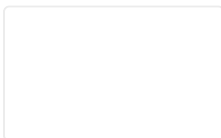


Caramel
#A9784B

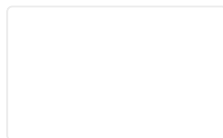


Tan
#C9A37A

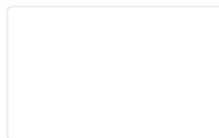
NEUTRALS



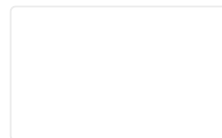
Charcoal
#2D2D2D



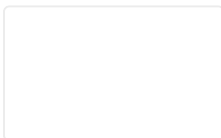
Dark Gray
#4F4F4F



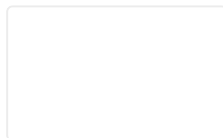
Medium Gray
#757575



Gray
#A0A0A0

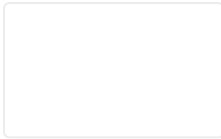


Light Gray
#C7C7C7

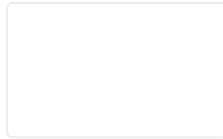


Off White
#EFEFEF

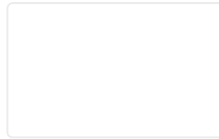
WARM NEUTRALS



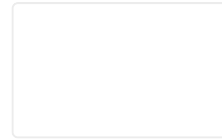
Mocha
#7A5A43



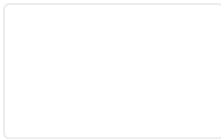
Coffee
#95724D



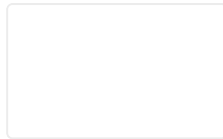
Camel
#B38B5E



Sand
#C8B28B

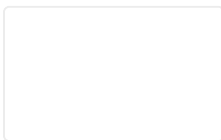


Oat
#DCC8A6

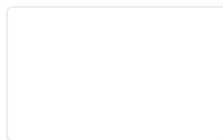


Bone
#E8DFCF

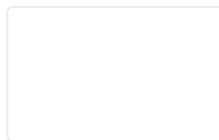
COOL GRAYS & SLATE



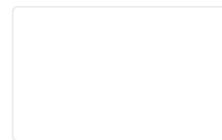
Slate
#4B5866



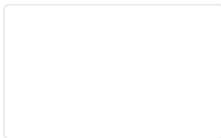
Blue Gray
#627386



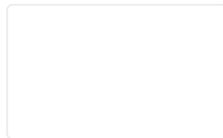
Steel Gray
#7D8A99



Stone
#9AA3AF

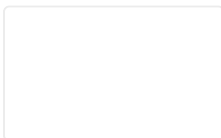


Cool Gray
#B5BAC3

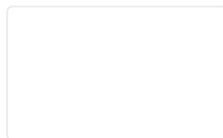


Fog
#CCD1D7

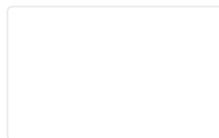
HIGHLIGHT PASTELS



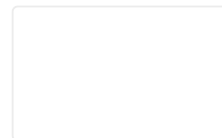
Pastel Peach
#F7CDBA



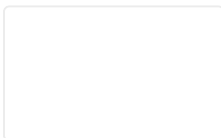
Pastel Apricot
#F8D9B5



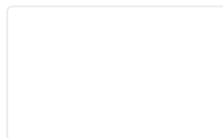
Pastel Yellow
#F9E6A6



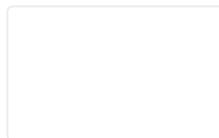
Pastel Mint
#BEE8D6



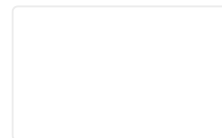
Pastel Aqua
#BCE7EA



Pastel Blue
#C7D9F3

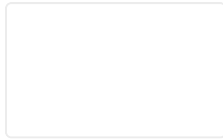


Pastel Lavender
#D6C9F2

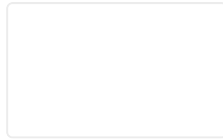


Pastel Pink
#F6C8E0

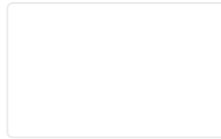
EARTH TONES



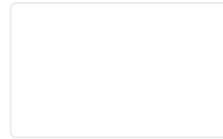
Olive Green
#606B3C



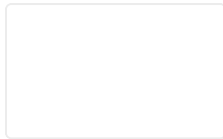
Army Green
#4E5730



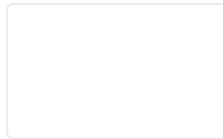
Moss Green
#6A7040



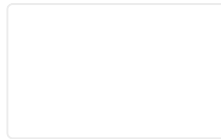
Rust
#A5522F



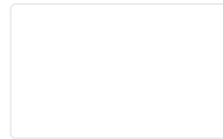
Terracotta
#C26E4A



Clay
#D2956A



Ochre
#D4AC5A



Beige
#E5D6B7

VIBRANT — CURATED FOR LIVELINESS OVER MAXIMUM LIFE

FOR THE PERSON WEARING IT

CARING FOR YOUR PRINT

Everything above is about the file and the production process. This part is the only piece in the customer's hands — and it matters just as much.

Wait 24–72 hours before the first wash.

That window lets the cure fully set before water and detergent test it for the first time.

Mild detergent. No bleach.

Bleach and harsh detergents break down the ink bond directly, not just the fabric around it.

Skip the fabric softener.

It coats fibers in a waxy film that dulls both the print and the fabric underneath it, wash over wash.

Wash inside-out, in cold water.

Cold water and reduced agitation against the print face slow down fading and surface wear.

Tumble dry low, or hang dry.

High heat in the dryer is a second, uncontrolled curing event — and not a gentle one.

GLOSSARY

PPI (pixels per inch) — the resolution of your design file, measured at the intended output size. This is what to specify when a printer asks for file resolution. Often incorrectly called DPI in everyday usage.

DPI (dots per inch) — the physical output resolution of the printer itself; a separate, much higher number than PPI. Your design file's PPI and the printer's DPI are different specifications.

Total ink coverage (TAC) — the sum of all ink channels at a given point, expressed as a percentage of maximum. 100% cyan + 100% magenta = 200% TAC. The ceiling varies by printer and RIP but typically runs 200–240% for light garments and 280–320% for dark garments including the white underbase.

Underbase — the white ink layer printed first on dark garments, giving color ink a visible surface to sit on. Without it, color ink deposited on dark fabric is invisible.

Pretreatment — a cationic polymer solution applied to fabric before printing. It reverses the fiber surface charge from negative to positive, causing the negatively charged pigment and binder in the ink to coagulate near the surface rather than wicking into the weave.

RIP (Raster Image Processor) — software that converts your design file into printer commands, handles color management, applies ICC profiles, enforces ink limits, and determines final ink volumes per pixel.

ICC profile — a file describing how a specific printer/ink/substrate combination reproduces color. Used by Photoshop and RIP software for soft proofing and accurate color conversion.

Crocking — ink rubbing off onto another surface on contact, either dry or wet. Wet crocking is more severe. Industry standard: AATCC Test Method 8.

Fibrillation — cotton fibers gradually breaking through the cured ink layer over time and washing, creating a worn or vintage look. Intrinsic to the pigment-on-fiber system.

Dye migration — disperse dyes in polyester fibers sublimating at curing temperatures and bleeding into the white ink layer, creating a color cast. Primary reason DTG results degrade on poly-cotton blends above ~50% polyester.

Metamerism — the phenomenon where two colors that appear identical under one light source look different under another. Caused by differences in the spectral reflectance of the pigment mixture.

Colorfastness — how well a color resists fading, bleeding, or shifting under washing, light exposure, and friction.

Gamut — the complete range of colors a device or process can reproduce. DTG's CMYK+white gamut is narrower than sRGB, with pronounced limitations in saturated blues, bright greens, and all neons.

Halftone — a pattern of flat, hard-edged dots at full ink density used to simulate continuous tones optically. Each dot is a binary decision — full ink or no ink at that point.

Curing — the heat process that cross-links the polymer binder around pigment particles, bonding them to the fiber surface. Defined by both temperature and dwell time specific to each ink/printer system.

DTG PRINTING: A FIELD GUIDE

A comprehensive guide to direct-to-garment printing color science and print longevity, compiled by Mumblejinx from production research, manufacturer documentation, and two original reference color palettes.

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v1.1 • July 2026 • Temperature ranges, ink coverage figures, and cure specifications reflect general industry practice across common DTG systems — always follow your specific printer and ink manufacturer's published specifications, and confirm against a physical proof before a production run.

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Provenance. The science in this guide is drawn from publicly available materials science, printer manufacturer documentation, and textile industry research — factual information that belongs to no one. Both color palettes were independently derived by Mumblejinx from that same underlying science, not copied or adapted from any third-party source. The synthesis, framing, and interactive tools are original work. Nothing here was paywalled, proprietary, or taken without the right to use it.